TEXORA · LECTURE SHEET

Historical Development of Looms

From a warp stretched between pegs to an air jet inserting weft at over a thousand picks a minute: the loom's history is a sequence of bottlenecks removed — first the slow hand-thrown shuttle, then human power itself, then the shuttle altogether.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 2 · LEVEL 2 TERM 1

1. Reading the chronology

Weaving is among the oldest technologies: interlacing warp and weft needs no power and no metal, so looms appear in almost every early civilisation. Yet for most of that history weaving stayed slow, because the loom's three basic actions — **shedding** (separating the warp to open a shed), **picking** (inserting one pick across the shed) and **beat-up** (pushing the pick into the cloth fell with the reed) — were all done by hand, one pick at a time. Every major development attacked one of those actions, or the stoppages between them: each invention survived because it removed the bottleneck then limiting production or quality.

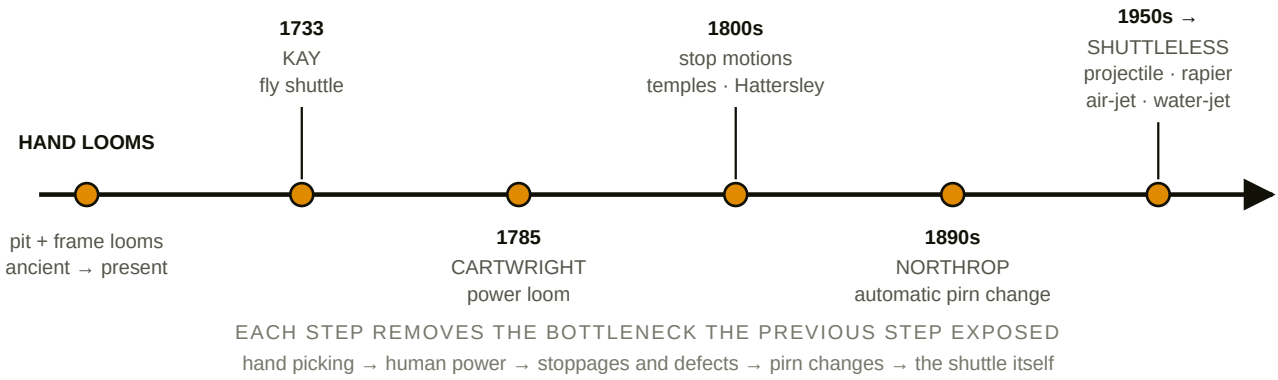


Figure 1 — Timeline of loom development. Landmarks: Kay's fly shuttle (1733), Cartwright's power loom (1785), 19th-century stop motions, temples and the Hattersley power loom, Northrop's automatic loom (1890s), and the shuttleless era from the 1950s.

2. Primitive and hand looms

The earliest looms simply held the warp under tension: the weaver separated the threads with fingers, a shed stick and a heddle rod, passed weft through by hand, and beat it in with a wooden sword or comb.

The cloth could be excellent, but output was measured in centimetres per day. Two hand-loom forms remain in use:

2.1 The pit loom

The **pit loom**, still used for traditional weaving in South Asia, stands over a pit dug in the floor. The treadles are worked by the weaver's feet inside the pit while the weaver sits at ground level at the cloth fell. Foot-worked shedding frees both hands for shuttle and reed — but picking and beat-up remain hand work, at the weaver's own pace.

2.2 The frame loom

The **frame loom** carries the warp on a rigid rectangular frame — small frames for samples, large treadle frame looms for full-width cloth. The developed handloom already has the modern arrangement: warp beam, shafts with healds worked by treadles, a lay carrying reed and shuttle boxes, and a cloth roller. It lacks only power and automation: the weaver throws the shuttle, beats up, advances warp and cloth, and stops for every weft fault. Such looms, with dobby or jacquard shedding, still weave fine, figured and heritage fabrics.

FEATURE	PIT LOOM	FRAME / TREADLE HANDLOOM
Warp support	Frame over a pit; warp at ground level	Rigid frame with warp beam and cloth roller
Shedding	Treadles worked by feet in the pit	Treadles and lamms under the frame
Picking	Shuttle thrown or passed by hand	Hand-thrown shuttle; fly shuttle on many later handlooms
Role today	Traditional cloths and saris	Craft, sampling, figured and heritage fabrics

3. The fly shuttle — Kay, 1733

The first bottleneck was picking: wide cloth once needed two weavers passing the shuttle between them, and the shuttle's pace set the loom's pace. In **1733 John Kay patented the fly shuttle**: the shuttle rests in a box at each end of the lay and is driven across the shed by a sharp blow from a **picker**, struck by a cord the weaver tugs — one tug sends it flying through the shed, a second returns it. One weaver could now weave any width, several times faster than by hand throwing. It also unbalanced the industry — weavers consumed weft faster than hand spinners could supply it — the first case of mechanising one stage exposing the next as the bottleneck.

4. The power loom — Cartwright, 1785

The next bottleneck was power itself. In **1785 Edmund Cartwright patented the first practical power loom**, driven from an external source through shafts and belts, its cranks and cams performing shedding, picking and beat-up in a fixed, timed cycle. Early power looms were crude and needed years of improvement by others before factory power weaving spread in the early 19th century. But the principle was set: the loom became a machine tended by an operative rather than driven by a weaver's muscle, running continuously at roughly 100–200 picks per minute.

5. Nineteenth-century refinements

Once a loom ran by power, its losses were stoppages and defects, and the 19th century added the protective and regulating motions now considered standard:

- **Warp stop motions.** Each end carries a drop wire; when an end breaks or slackens the wire falls and stops the loom before the fault is woven in. Hand looms had no such protection.
- **Weft stop motions (weft forks).** A feeler checks that a pick was inserted; if the weft breaks or the pirn empties, the loom stops instead of weaving cloth with missing picks.
- **Temples.** A temple grips each cloth edge near the fell — a ring or roller set with fine pins — holding the fabric out to nearly its reed width. Without temples the cloth contracts as the weft crimps, edge ends are dragged inward and abraded by the reed, and edge breaks follow.
- **Regulating motions.** Improved let-off and take-up kept warp tension and pick density regular; guards and check motions controlled fly-shuttle picking at rising speeds.

5.1 The Hattersley loom

George Hattersley of Keighley, Yorkshire, became a defining loom maker of this period, and **Hattersley loom** is still the name used in many regions for the firm's well-built, fast plain power looms, produced from the mid-19th century onward, including dobby-fitted versions. The type represents the mature shuttle power loom: timed basic motions, warp and weft stop motions, temples, and speeds around 180–220 picks per minute — the standard factory loom against which later looms were judged.

6. The automatic loom — Northrop, 1890s

A power loom stopped every time a pirn emptied — every few minutes at power-loom speeds — so the weaver's attention, not picking speed, now limited how many looms one weaver could tend. The answer, developed in the **1890s** by **James H. Northrop** and the Draper company, was the **automatic loom**: a rotating magazine, the Northrop battery, holds full pirns above the shuttle box, and when a feeler finds the running pirn nearly empty, a fresh pirn is pressed into the shuttle *while the loom keeps running* and the empty pirn ejected. With stop motions fitted, such looms ran long periods unattended and one weaver could tend many looms — weaving's first large jump in labour productivity. Two

families exist: **pirn-changing** (Northrop type — a new pirn into the same shuttle) and **shuttle-changing** (a loaded shuttle replaces the exhausted one). Pirn-changing automatics dominated cotton weaving through much of the 20th century.

7. Semi-automatic looms

Between the plain power loom and the full automatic stands the **semi-automatic loom**: a power loom with *some* automatic devices — commonly warp and weft stop motions — but no automatic pirn or shuttle change, so the weaver still stops the loom to recharge the shuttle. Such looms suited varied or short work where full automation's cost could not be repaid; many Hattersley-type looms are of this class. The test is weft replenishment: renewal *without stopping* is automatic; stopping to recharge is semi-automatic if protective motions are fitted, otherwise non-automatic.

8. The shuttleless era

By the mid-20th century the shuttle itself was the bottleneck: accelerating it across the shed and arresting it in the opposite box consumes most of the picking energy, makes great noise, limits width and speed, and its small pirn empties quickly. **Shuttleless looms** carry only the weft across, fed from a large stationary cone — so pirn winding (Topic 1) and replenishment stoppages vanish as well.

SYSTEM	HOW THE PICK IS INSERTED	CHARACTER
Projectile — Sulzer, commercialised in the 1950s	A small gripper projectile, about 90 mm long, is shot across the shed carrying the weft tip, then returned empty under the warp.	First commercially successful shuttleless system; very wide widths, gentle weft handling, versatile
Rapier — widespread from the 1960s–70s	Rigid rods or flexible tapes carry the weft to mid-shed and hand it over, or one rapier carries it right across.	Most versatile: almost any weft, many colours, delicate yarns
Air-jet — dominant for volume work by the late 20th century	A main nozzle blows the pre-measured weft across on a timed air jet, assisted by relay nozzles along a profiled reed.	Highest insertion rates — over 1000 picks/min; needs even weft, consumes compressed air
Water-jet — late 20th century	A jet of water carries the weft across; loom and cloth are wet during weaving.	Very fast and quiet for hydrophobic filament yarns (polyester, nylon); unsuitable for yarns swollen or weakened by water

Shuttleless looms raised the **weft insertion rate** — picks per minute × working width — several-fold. Their costs are capital price and, for air-jet, compressed air; their demand is a well-prepared warp, because at 800–1500 picks/min no weak end is forgiven.

9. Modern electronic looms

The latest stage replaces mechanical timing with electronic control. Electronic let-off and take-up hold warp tension and pick density to set values continuously; electronic dobby and jacquard shedding store patterns as data, changed in minutes; every stop is recorded with its cause. Servo drives remove belts and gears, letting each motion be optimised in software. Air-jet looms now exceed 1200 picks/min in production — but the deeper change is control: the modern loom reports and adjusts its own work.

> **WORKED EXAMPLE — WHAT SHUTTLELESS SPEED MEANS IN CLOTH**
>
> Fabric construction: 40 picks/inch.
>
> Picks per metre = 40 × 39.3701 = **1574.8 picks/m** (1 m = 39.3701 in).
>
> **Shuttle loom at 180 picks/min:** picks per hour = 180 × 60 = 10,800; cloth output = 10,800 ÷ 1574.8 ≈ **6.86 m/h**.
>
> **Air-jet loom at 1200 picks/min:** picks per hour = 1200 × 60 = 72,000; cloth output = 72,000 ÷ 1574.8 ≈ **45.7 m/h**.
>
> Ratio = 45.7 ÷ 6.86 ≈ **6.7×** — equivalently 1200 ÷ 180 = 6.67, since at the same pick density outputs follow the speeds. Both figures are theoretical (100 % efficiency).

10. Classification of looms

Looms are classified on several independent axes; a full description uses one term from each.

10.1 The three main axes

By source of power: hand looms — all motions by the weaver (pit, frame, treadle handlooms); **power looms** — motions driven by an external source, from Cartwright's loom to the modern electronic loom.

By weft replenishment (automaticity): non-automatic — the loom stops and the weaver recharges the shuttle when the pirn empties; **semi-automatic** — power loom with protective motions but manual replenishment; **automatic** — weft renewed without stopping (pirn-changing, shuttle-changing, or continuous cone feed on shuttleless looms).

By weft-insertion system: shuttle looms — a shuttle carrying a pirn traverses the shed (hand, plain power and automatic shuttle looms); **shuttleless looms** — projectile, rapier (rigid or flexible, single or double), air-jet and water-jet looms.

Further axes: **by shedding device** (tappet, dobby, jacquard) and **by fabric** (light, medium, heavy, terry, leno, carpet, narrow fabrics).

EXAMPLE LOOM	POWER	AUTOMATICITY	INSERTION
Pit loom with fly shuttle	Hand	Non-automatic	Shuttle
Hattersley-type loom, stop motions, hand pirn change	Power	Semi-automatic	Shuttle
Northrop loom	Power	Automatic (pirn-changing)	Shuttle
Sulzer projectile / modern air-jet loom	Power	Automatic (cone supply)	Shuttleless

Glossary

Shed — the opening between raised and lowered warp ends for the pick to pass. **Pick** — one weft insertion, and the inserted yarn itself. **Beat-up** — pushing the new pick into the cloth fell with the reed. **Fly shuttle** — Kay's shuttle (1733), driven by pickers and a cord. **Power loom** — a loom driven by external power; Cartwright's patent, 1785. **Warp stop motion** — device stopping the loom when an end breaks, usually through a drop wire. **Temple** — device holding the cloth out to width at the fell. **Automatic loom** — a loom renewing its weft supply without stopping. **Northrop battery** — the rotating pirn magazine of the Northrop loom. **Shuttleless loom** — a loom inserting weft by projectile, rapier, air or water jet. **Weft insertion rate** — picks/min × working width.

Key points

• Loom history is a chain of bottlenecks: hand picking → human power → stoppages and defects → pirn changing → the shuttle itself → control.
• Pit and frame hand looms already contain the modern arrangement of beam, shafts, reed and cloth roller; they lack power and automation.
• Kay's fly shuttle (1733) mechanised picking; Cartwright's power loom (1785) mechanised the whole timed cycle.
• The 19th century made the power loom safe to leave running: stop motions, temples, regulated let-off and take-up; Hattersley looms typify it.
• The Northrop automatic loom (1890s) renewed the pirn without stopping, multiplying the looms one weaver could tend; semi-automatic looms have stop motions but manual replenishment.
• Shuttleless systems (Sulzer projectile, 1950s; then rapier, air-jet, water-jet) removed shuttle and pirn; electronic looms add data-driven control.

Practice

1. Place in chronological order, with date or period: Cartwright's power loom; Kay's fly shuttle; the Northrop automatic loom; the Sulzer projectile loom; temples and warp stop motions.
2. A fabric has 50 picks/inch and a loom runs at 220 picks/min. Calculate the theoretical cloth output in m/h, showing every step.
3. Classify on all three main axes: (a) a treadle handloom with fly shuttle; (b) a power loom with stop motions whose shuttle the weaver recharges by hand; (c) an air-jet loom fed from cones.
4. Why did the fly shuttle create a crisis in spinning, and which later invention shows the same bottleneck pattern?
5. A mill weaves hydrophobic polyester filament and wants the fastest insertion. Which shuttleless system is the classic choice, and which is ruled out for cotton — for opposite reasons involving water?
6. Explain why the shuttle became the limiting element by the mid-20th century (energy, noise, width, weft supply).

Answer key

1. Kay's fly shuttle — **1733**; Cartwright's power loom — **1785**; stop motions and temples — **1800s**; Northrop automatic — **1890s**; Sulzer projectile — **1950s**.
2. Picks per metre = 50 × 39.3701 = 1968.5 picks/m. Picks per hour = 220 × 60 = 13,200. Output = 13,200 ÷ 1968.5 ≈ **6.71 m/h**, theoretical at 100 % efficiency.
3. (a) Hand · non-automatic · shuttle. (b) Power · semi-automatic · shuttle. (c) Power · automatic (continuous cone supply) · shuttleless (air-jet).
4. One weaver could suddenly weave wider cloth several times faster, so weft consumption outran hand spinning and yarn ran short: mechanising one stage transfers the bottleneck to an adjoining stage — as later with pirn changing (answered by the Northrop automatic) and the shuttle itself (answered by shuttleless insertion).
5. The **water-jet loom**: water does not damage hydrophobic polyester filament, and insertion is among the fastest. Ruled out for cotton is the same water-jet, for the opposite reason: cotton absorbs water and swells, and the cloth would leave the loom soaked.
6. Energy: the shuttle is far heavier than its weft, yet is accelerated and arrested every pick, so energy and wear grow with speed and width. Noise: its flight and checking dominate loom noise. Width: flight time limits practical width. Weft supply: the internal pirn holds only a short length.